Visualizing Urban Heat Island Effect in Las Vegas

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Powered by ACIS Regional Climate Centers

Background:

As climate change is raising temperatures all over the globe, cities are being disproportionately effected by these rises. In comparison to cities' rural areas, the rapid rise of average temperatures is striking. This is known as the Urban Heat Island Effect (Environmental Protection Agency).

Data Collection:

Using inspiration from Climate Central's methodology, I chose four rural weather stations around a city to the north, south, east and west. The stations had to match three criteria to qualify as a rural location: 1) be between 50 and 150 miles from the city, 2) lie in a population area of less than 10,000, and 3) lie in dim, dark, or unlit land areas at night. See the image to the right for a night satellite map of the city of Las Vegas (red) and the four rural locations surrounding (green).

Then, using Dark Sky API and ACIS Regional Climate Centers, I collected all available daily temperature data from each station.

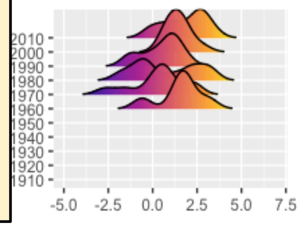
Data Manipulation:

Daily station data from decades ago was often missing. To handle this missing data, I used interpolation to estimate temperatures using Amelia R package. To manipulate data for creating visualization, I used dpylr as well as other R packages.

Average Temperature Anomaly Distrubutions Each Decade

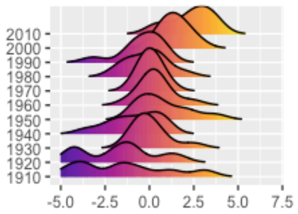
This visualization shows distributions of temperature anomalies. A temp. anomaly is the difference of an observed temp. from a baseline. The distributions are of yearly temp. anomalies over each decade.

North

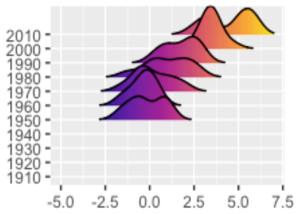


The baseline used for measuring anomalies for each location was calculated by taking the average of the middle 50% of data available. All temperature is measured in Fahrenheit.

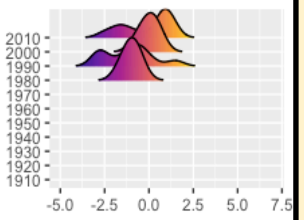
West



City of Las Vegas

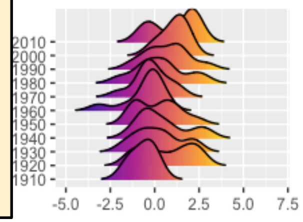


East



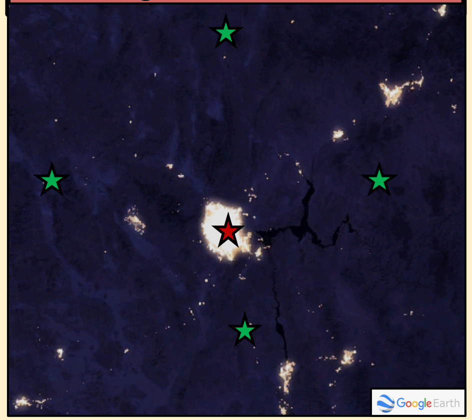
The purple hue represents a negative anomaly, indicating the measured temperature was below the the baseline temperature. The orange hue represents a positive anomaly, indicating a rise in temp. from the baseline.

South



In all five locations, there is a trend of increase in temperature anomalies as time increases. However, in the City this trend is much more extreme, shown by the rightward motion of the ridges

Las Vegas and rural stations



Tools Used:

R packages:

Amelia, ggplot, dplyr, gganimate, plotly. gridExtra, ggridges

Python libraries:

Plotly, numpy, pandas

Other:

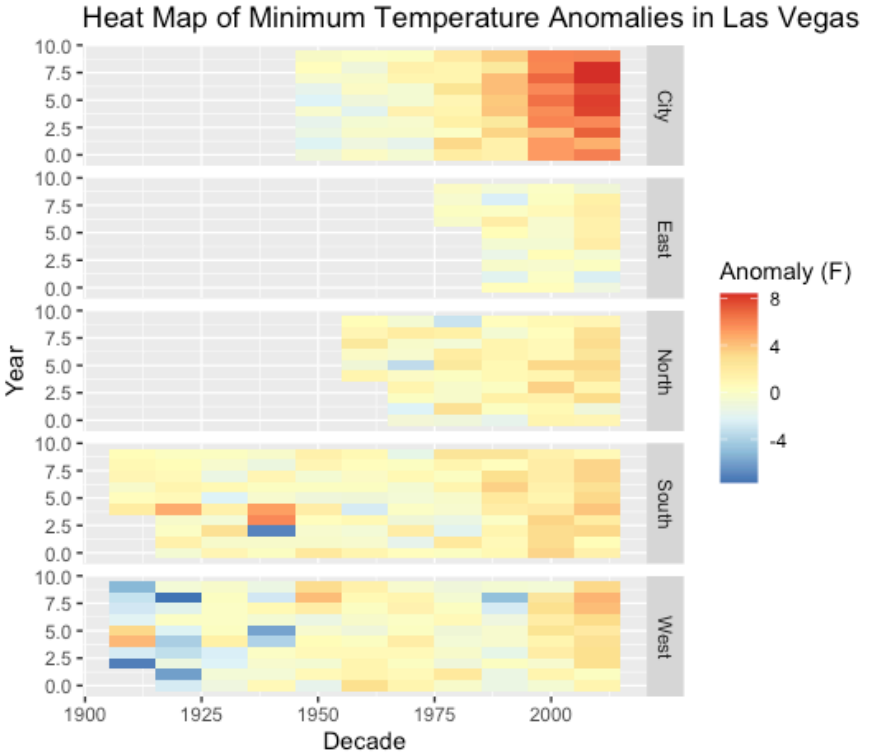
DarkSky API, Northeast RCC CLIMOD 2, Google Earth's Night on Earth

Citations:

Alyson Kenward PhD, Daniel Yawitz, Todd Sanford PhD, and Regina Wang (2014). "Summer in the City: Hot and Getting Hotter." *Climate Central*. http://assets.climatecentral.org/pdfs/UrbanHeatIsland.pdf

Environmental Protection Agency. "Heat Island Effect." *EPA*, Environmental Protection Agency, 23 Apr. 2020, www.epa.gov/heat-islands.

Daily minimum temperatures, often at night, are rising in cities much faster than in surrounding rural areas. The graph below shows the anomalies of minimum temperatures in each location.



Conclusions:

In addition to studying Las Vegas, I also studied San Francisco and Minneapolis. Both urban areas showed similar trends as Las Vegas. Scan QR code to the left to see more visualizations. These visualizations can easily be expanded to any city.

This type of work is important because it helps the average person visualize the impact climate change is having on urban centers. U.N. estimates 55%, or more than 4 billion people, live in urban areas around the world. According to the U.S. Census Bureau, 80% of Americans live in urban areas. The UHIE in urban areas are negatively impacting billions of people and will continue to as cities and populations grow.